

SAFETY DATA SHEET

Carbon Dioxide, Compressed

SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

1.1 Product Details

Product Name : Carbon Dioxide
Recommended use : Industrial and professional use

1.2 Company Identification

Company's Name : Leeden Gases Sdn. Bhd.
Company's Address : Lot PT 5074 & 5075, Jalan Jangur 28/43,
Seksyen 28, 40400 Shah Alam, Selangor.
Emergency telephone Number : 03-55228222 (Hunting Line) / 03-55228288 (Hotline)

SECTION 2: HAZARD IDENTIFICATION

OSHA regulatory status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).

Hazard classification : GASES UNDER PRESSURE - Liquefied gas.
Simple asphyxiant.

Hazard pictogram :



Signal word : Warning.

Hazard statements :

- Contains gas under pressure; may explode if heated.
- May displace oxygen and cause rapid suffocation.

Precautionary statements:

Storage : Protect from sunlight. Store in a well-ventilated place.

Disposal : Dispose of contents/ container in accordance with local/ regional/ national/ international regulations.

Other hazard : May cause frostbite upon sudden release of liquefied gas.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Chemical name : Carbon Dioxide
CAS number : 124-38-9
Proportion (%) : 100.0%

SECTION 4: FIRST AID MEASURES

4.1 Description of necessary first aid measures

Eye contact : Contact with liquid: Immediately flush eyes with plenty of water for at least 15 minutes. Then get immediate medical attention.

Inhalation : If adverse effects occur, remove to uncontaminated area. Give artificial respiration if not breathing. If breathing is difficult, oxygen should be administered by qualified personnel. Get immediate medical attention.

Skin contact : If frostbite or freezing occur, immediately flush with plenty of lukewarm water (105-115°F; 41-46°C). DO NOT USE HOT WATER. If warm water is not available, gently wrap affected parts in blankets. Get immediate medical attention.

Ingestion : If swallowed, get medical attention.

Most Important Symptoms/Effects

Acute

Frostbite, suffocation, Depression of central nervous system.

Note to Physicians

For inhalation, consider oxygen.

SECTION 5: FIRE FIGHTING MEASURES

Extinguishing Media : Use extinguishing agents appropriate for surrounding fire.

Special Hazards Arising from the Chemical : Negligible fire hazard. Containers may rupture or explode if exposed to heat.

Hazardous Combustion Products : Oxides of carbon.

Fire Fighting Measures : Move container from fire area if it can be done without risk. Cool containers with water spray until well after the fire is out. Stay away from the ends of tanks. For tank, rail car or tank truck, evacuation radius: 800 meters (1/2 mile). Use extinguishing agents appropriate for surrounding fire. Cool containers with water spray until well after the fire is out. Do not get water directly on material. Avoid inhalation of material or combustion by-products. Stay

Special Protective Equipment and Precautions for Firefighters : upwind and keep out of low areas.
: Wear personal protective clothing and equipment such as self-contained breathing apparatus (SCBA) for protection against possible exposure.

SECTION 6: ACCIDENTAL RELEASE MEASURE

Personal Precautions, Protective Equipment and Emergency Procedures

Wear personal protective clothing and equipment, see Section 8.

Methods and Materials for Containment and Cleaning Up

Leaking gas fire: Do not extinguish, unless leak can be stopped safely. Avoid heat, flames, sparks and other sources of ignition. Stop leak if possible without personal risk. Reduce vapors with water spray. Keep unnecessary people away, isolate hazard area and deny entry. Remove sources of ignition. Ventilate closed spaces before entering. Notify Local Emergency Planning Committee and State Emergency Response Commission for release greater than or equal to RQ (U.S. SARA Section 304).

Environmental Precautions

Avoid release to the environment.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Use only outdoors or in a well-ventilated area. Avoid breathing dust/fume/gas/mist/vapors/spray.

Conditions for Safe Storage, Including any Incompatibilities

Protect from sunlight. Store in a well-ventilated place. Store and handle in accordance with all current regulations and standards. U.S. OSHA 29 CFR 1910.101.

Incompatible Materials

Combustible materials, oxidizing materials, metal salts, reducing agents, metal carbide, metals, bases, potassium, sodium, ethyleneimine

SECTION 8: EXPOSURE CONTROL AND PERSONAL PREOTECTION

Component Exposure Limits

ACGIH:	5000 ppm TWA
	30000 ppm STEL
NIOSH:	5000 ppm TWA ; 9000 mg/m ³ TWA
	30000 ppm STEL ; 54000 mg/m ³ STEL
	40000 ppm IDLH
Europe:	5000 ppm TWA ; 9000 mg/m ³ TWA
OSHA (US):	5000 ppm TWA ; 9000 mg/m ³ TWA
Mexico:	5000 ppm TWA VLE-PPT ; 9000 mg/m ³ TWA VLE-PPT
	15000 ppm STEL [PPT-CT] ; 27000 mg/m ³ STEL [PPT-CT]

Engineering Controls

Provide local exhaust ventilation system. Ensure compliance with applicable exposure limits.

Individual Protection Measures, such as Personal Protective Equipment

Eye/face protection

For the gas: Eye protection not required, but recommended. For the liquid: Wear splash resistant safety goggles. Contact lenses should not be worn. Provide an emergency eye wash fountain and quick drench shower in the immediate work area.

Skin Protection

For the gas: Protective clothing is not required. For the liquid: Wear appropriate protective, cold insulating clothing.

Respiratory Protection

Any self-contained breathing apparatus that has a full facepiece and is operated in a pressure-demand or other positive-pressure mode.

Glove Recommendations

For the gas: Protective clothing is not required. For the liquid: Wear appropriate protective, cold insulating clothing.

SECTION 9: STABILITY AND REACTIVITY

Physical state	: Gas.
Colour	: Colourless.
Odor	: Odorless
pH	: (Acidic in solution)
Melting Point	: -57 °C at 4000 mmHg (-71 °F)
Boiling Point	: -78.5 - -61.7 °C (-109 - -79 °F Liquid)
Vapor Pressure	: 43700 mmHg @ 21 °C
Vapor Density (air=1)	: 1.5
Specific Gravity (water=1)	: 1.527 at 21 °C

Viscosity	: 2.03E-06 Pa.s
Sublimation	: -78.5 °C (-109 °F)
Taste	: Acid taste
Volatility by Volume	: 100%
Molecular formula	: CO ₂
Molecular weight	: 44.01 g/mole
Triple point	: -56.6 °C

Soluble

Alcohol, acetone, Hydrocarbons, organic solvents

SECTION 10: STABILITY AND REACTIVITY

Reactivity	: No reactivity hazard is expected.
Chemical stability	: Stable at normal temperatures and pressure.
Possibility of hazardous reactions	: Will not polymerize.
Conditions to Avoid	: Protect from physical damage and heat. Containers may rupture or explode if exposed to heat. Avoid contact with water or moisture.
Hazardous decomposition products	: Oxides of carbon
Hazardous polymerization	: Combustible materials, oxidizing materials, metal salts, reducing agents, metal carbide, metals, bases, potassium, sodium, ethyleneimine

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Likely Routes of Exposure

Inhalation

Changes in blood pressure, ringing in the ears, nausea, difficulty breathing, irregular heartbeat, headache, drowsiness, dizziness, tingling sensation, tremors, weakness, visual disturbances, suffocation, convulsions, Unconsciousness, coma

Skin Contact

Blisters, frostbite

Eye Contact

Frostbite, blurred vision

Ingestion

Ingestion of a gas is unlikely

Acute and Chronic Toxicity**Component Analysis - LD50/LC50**

The components of this material have been reviewed in various sources and no selected endpoints have been identified.

Product Toxicity Data**Acute Toxicity Estimate**

No data available.

Immediate Effects

Frostbite, suffocation, Depression of central nervous system.

Medical Conditions Aggravated by Exposure

Heart or cardiovascular disorders, respiratory disorders

SECTION 12: ECOLOGICAL INFORMATION

No known toxicological effects from this product.

SECTION 13: DISPOSAL INFORMATION

Disposal method : Dispose in accordance with all applicable regulations.

SECTION 14: TRANSPORT INFORMATION

UN Number : UN 1013
UN proper shipping name : Carbon Dioxide
Hazard class / Division : 2.2 (Non-flammable gas)



SECTION 15: REGULATORY INFORMATION

National Regulations : Occupational Safety and Health Act 1994 and relevant regulations:

- Occupational Safety and Health (Classification, Labeling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013.
- Occupational Safety and Health (Use And Standards of Exposure of Chemicals Hazardous to Health) Regulations 2000

Environment Quality Act 1974 & regulations:

- Environment Quality (Clean Air) Regulations 2014
- Environmental Quality (Scheduled Wastes) Regulations 2005

SECTION 16: OTHER INFORMATION

NFPA Ratings

Health: 2 Fire: 0 Reactivity: 0 Other: SA

Hazard Scale: 0 = Minimal 1 = Slight 2 = Moderate 3 = Serious 4 = Severe